

73. $\phi'(x) = -\frac{2xy+12y^3x^3}{x^2+9y^2x^4}$

74. $\frac{\partial \phi}{\partial y}(y, z) = -\frac{2x^2y}{2xy^2 + e^x}$ e $\frac{\partial \phi}{\partial z}(y, z) = -\frac{1}{2xy^2 + e^x}$

75. a. ϕ deverá ser de classe $\mathcal{C}^1$ numa vizinhança de $\frac{y_0}{z_0}$ e $\phi'(\frac{y_0}{z_0}) \neq \frac{x_0}{y_0}$

76. b. $z'_u(1, \frac{\pi}{2}) = -\frac{2\pi+4}{\pi+4}$

77. a. $\frac{\partial^2 \phi}{\partial x^2}(0, 0) = 0$

b. $\phi(x, z) = \arcsin(-x) - z$ e $\frac{\partial^2 \phi}{\partial x^2}(0, 0) = 0$

78. $y'(x) = -\frac{5x}{y}$ e $z'(x) = -\frac{4x}{z}$

79.

$$\begin{aligned} y'(x) &= \frac{(-e^x \cos(yz) - 2x)(\cos(xyz)xy + 2z) - e^x \sin(yz)y(\cos(xyz)yz + 2x)}{-e^x \sin(yz)z(\cos(xyz)xy + 2z) + e^x \sin(yz) \cos(xyz)xyz} = \\ &= \frac{(-e^x \cos(yz) - 2x)(\cos(xyz)xy + 2z) - e^x \sin(yz)y(\cos(xyz)yz + 2x)}{-e^x \sin(yz)2z^2} \end{aligned}$$

e

$$\begin{aligned} z'(x) &= \frac{e^x \sin(yz)z(\cos(xyz)yz + 2x) + \cos(xyz)xz(e^x \cos(yz) + 2x)}{-e^x \sin(yz)z(\cos(xyz)xy + 2z) + e^x \sin(yz) \cos(xyz)xyz} = \\ &= \frac{e^x \sin(yz)z(\cos(xyz)yz + 2x) + \cos(xyz)xz(e^x \cos(yz) + 2x)}{-e^x \sin(yz)2z^2} \end{aligned}$$

81. b. $Jac f^{-1}(u, v) = -\frac{1}{2e^{x+y}} \begin{bmatrix} -e^y & -e^y \\ -e^x & e^x \end{bmatrix}$, com $f(x, y) = (u, v)$

82. f é invertível numa vizinhança de $(\frac{1}{2}, \frac{1}{2})$